Translations of tangent functions

1 Which of the following is the correct domain of $y = \tan x$?

A All real numbers, except integer multiples of 90°.

B All real numbers, except odd integer multiples of 90°.

C All real numbers, except even integer multiples of 90°. **D** All real numbers.

2 How has the graph $y = \tan x + 4$ been transformed from $y = \tan x$?

3 State whether the following statement is true or false:
The graph of $y = \tan\left(x + \frac{\pi}{4}\right)$ has a horizontal translation to the left from $y = \tan x$.

4 Consider the function $y = \tan(x - 90^\circ)$.

a Does the function have y-intercept?

b Determine the period of the function.

c How far apart are the asymptotes of the function?

d State the first asymptote of the function for $x > 0$.

e State the first asymptote of the function for $x \leq 0$.

f Sketch the graph of the function.

5 Consider the function $y = \tan(x - 60^\circ)$.

a Determine the value of y at the y-intercept.

b Determine the period of the function.

c How far apart are the asymptotes of the function?

d State the first asymptote of the function for $x \geq 0$.

e State the first asymptote of the function for $x \leq 0$.

f Sketch the graph the function.

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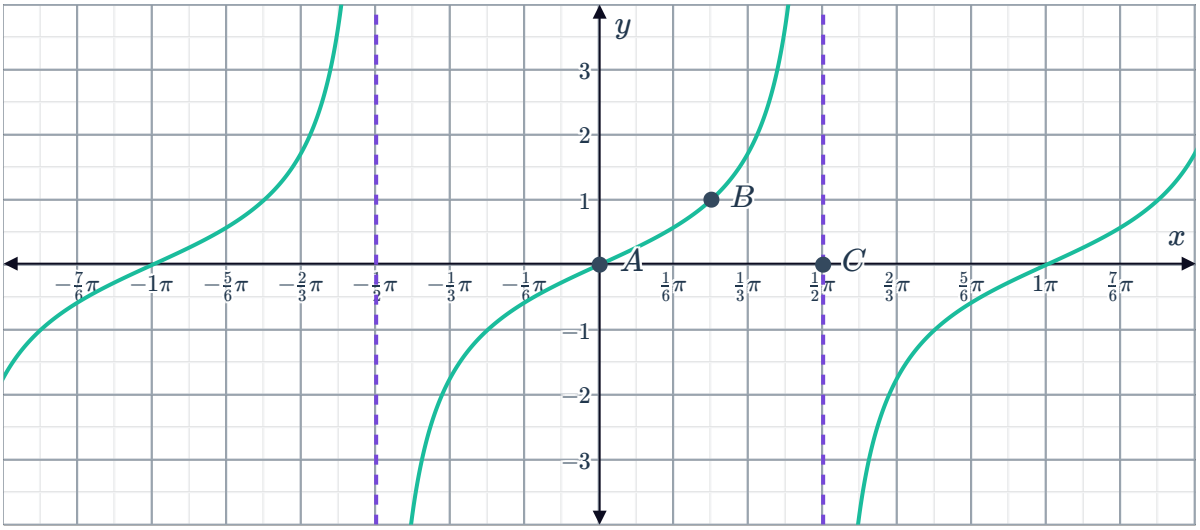
6 The function $f(x) = \tan\left(x - \frac{\pi}{6}\right)$ is to be graphed on the interval $\left[\frac{2\pi}{3}, \frac{8\pi}{3}\right]$.

a Find the equations of the asymptotes of the function that occur on this interval.

b Find the x-intercepts of the function that occur on this interval.

c Sketch the function $f(x)$ on the interval.

7 Consider the graph of $f(x) = \tan x$ and three points $A(0,0)$, $B\left(\frac{\pi}{4}, 1\right)$ and $C\left(\frac{\pi}{2}, 0\right)$:



a If $f(x)$ undergoes a transformation to $g(x) = \tan\left(x - \frac{\pi}{3}\right)$, state the coordinates of the following points after the transformation:

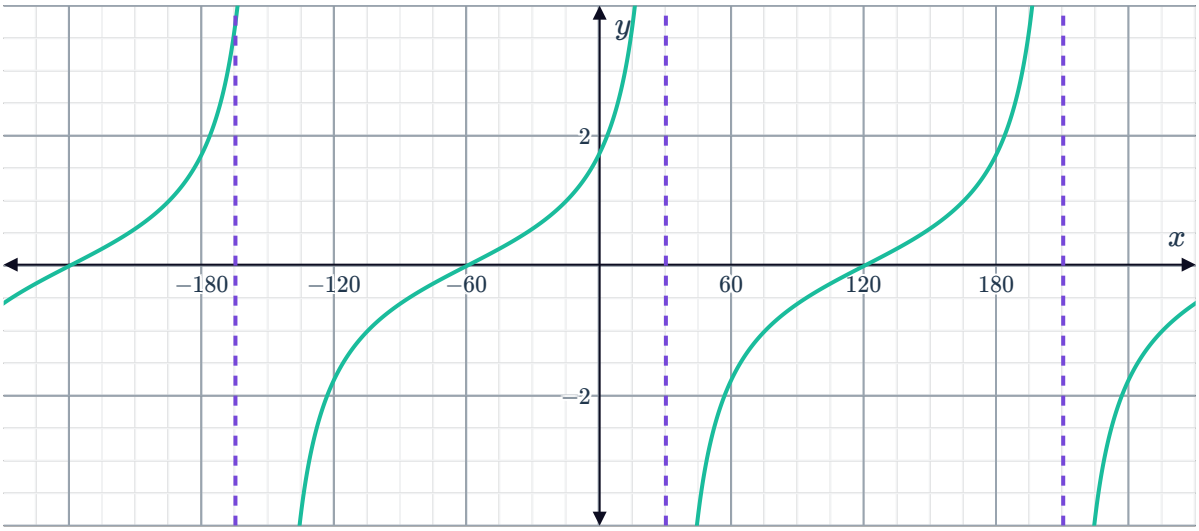
i A ii B iii C

b Describe the transformation of $f(x)$ to $g(x)$.

c Now, sketch the graph of $g(x)$ on the domain $-\pi \leq x \leq \pi$.

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8 Consider the graph shown:



a Determine whether the following could be the equation of the graph:

i $p(x) = \tan(x - 120^\circ)$

ii $q(x) = \tan(x + 60^\circ)$

iii $r(x) = \tan(x - 1020^\circ)$

iv $s(x) = \tan(x - 680^\circ)$

b Now, which of the following statements is true?

A The graph of $y = \tan(x - h)$ is the same as the graph of $y = \tan(x - h - 180n)$, for any integer n.

B The graph of $y = \tan(x - h)$ can be reflected in the y-axis to make the graph of $y = \tan(x - h - 180n)$, for any integer n.

C The graph of $y = \tan(x - h)$ is different for every value of h.

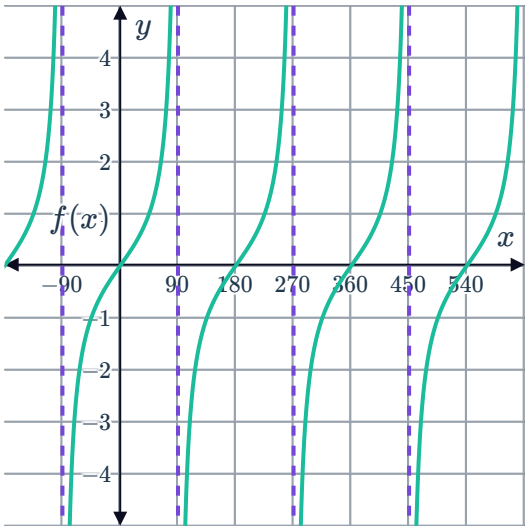
D The graph of $y = \tan(x - h)$ is the same as the graph of $y = \tan(x - h - 180n)$ only when $n = \pm 1$.

9 Let $f(x) = \tan x$ and $g(x) = \tan(x - 45^\circ)$.
What can be said of the domain and range of both functions? Explain your answer.

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Dilations of tangent functions

10 The graph of $f(x) = \tan x$ is shown.
On the same coordinate plane, sketch the graph of $g(x) = \tan\left(\frac{1}{2}x\right)$.



11 Consider functions of the form $y = \tan bx$.

a Complete the table, identifying the period of the function when $b = 1, 2, 3, 4$.

b State the period of $y = \tan bx$.

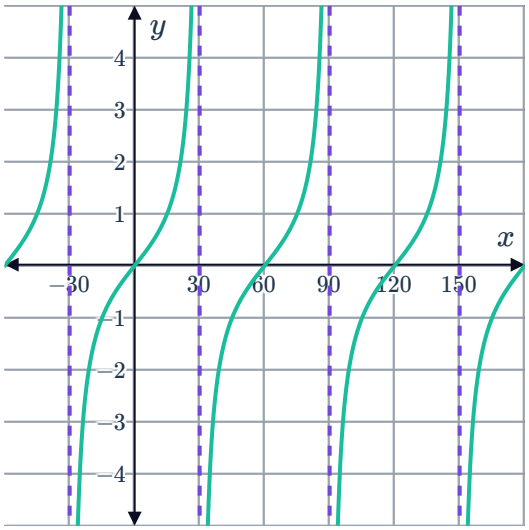
c As the value of b increases, describe the effect on the period of $y = \tan bx$.

Function	Period
$y = \tan x$	π
$y = \tan 2x$	
$y = \tan 3x$	
$y = \tan 4x$	

12 Consider the graph of a function in the form $f(x) = \tan bx$:

a State the period of the function.

b Now, state the equation of the function.



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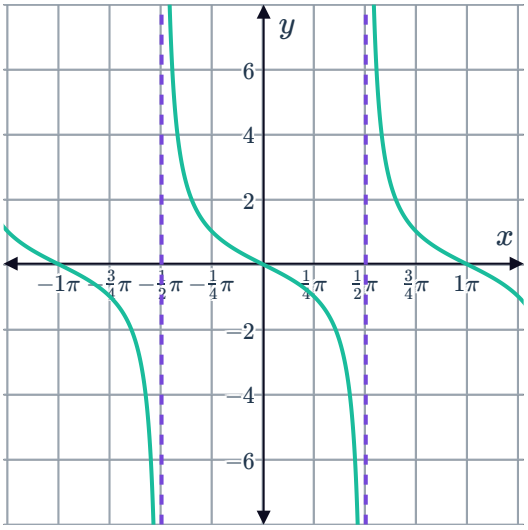
13 An asymptote of a function of the form $g(x) = \tan bx$ is $x = \frac{\pi}{8}$. Find the equation of $g(x)$.

14 How has the graph $y = 3\tan x$ been transformed from $y = \tan x$?

15 On the same coordinate plane, sketch the graphs of $y = \tan x$ and $y = \frac{1}{2}\tan x$, on the domain $-2\pi \leq x \leq 2\pi$.

16 Describe how the graph of $y = -5\tan x$ has been transformed from the function $y = \tan x$.

17 Consider the graph of the given tangent function:



- **a** The given graph is the result of what transformation from $y = \tan x$?
- **b** State the equation of the graphed function.

18 For each of the following functions:

- **i** Determine the y-intercept.
- **ii** Determine the interval between the vertical asymptotes of the function.
- **iii** Now, state the period of the function.
- **iv** State the equation of the first asymptote of the function for $x \geq 0$.
- **v** State the equation of the first asymptote of the function for $x \leq 0$.
- **vi** Sketch the graph the function on the domain $-\pi \leq x \leq \pi$.

a $y = -\tan x$

b $y = \tan\left(x + \frac{\pi}{3}\right)$

c $y = \tan\left(\frac{x}{2}\right)$

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Series of transformations

19 For each of the following functions:

- **i** Find the value of y when $x = \frac{\pi}{4}$.
- **ii** Determine the period of the function.
- **iii** Now, state the interval between the asymptotes of the function.
- **iv** State the equation of the first asymptote of the function for $x \geq 0$.
- **v** State the equation of the first asymptote of the function for $x \leq 0$.
- **vi** Sketch the graph the function on the domain $-\pi \leq x \leq \pi$.

a $y = 5\tan x + 3$

b $y = 4\tan 3x$

20 Consider the function $y = \tan\left(2x + \frac{\pi}{3}\right)$:

a Rewrite the function so that it is in the form $y = \tan\left(a\left(x + b\right)\right)$.

b Now, sketch the graph of $y = \tan\left(2\left(x + \frac{\pi}{6}\right)\right)$.

21 Sketch the graph of the following functions:

a $y = \tan\left(2\left(x - 45\right)\right)$

b $y = \tan\left(2x + 90^\circ\right)$

22 Consider the graphs of the following functions and state whether or not the graph is the same as $y = -\tan x$:

a $y = -\tan\left(x + \frac{3\pi}{4}\right)$

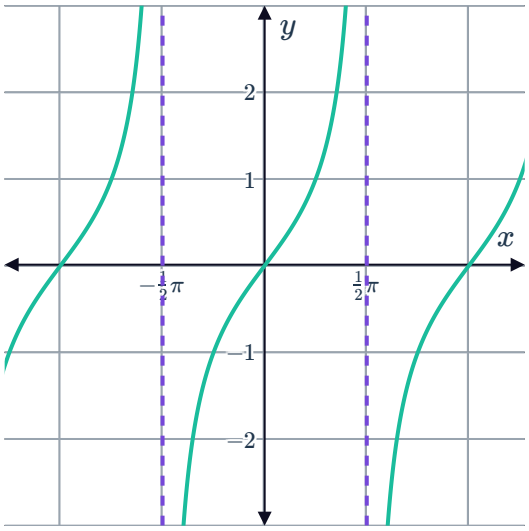
b $y = -\tan\left(x + \pi\right)$

c $y = -\tan\left(x + \frac{\pi}{2}\right)$

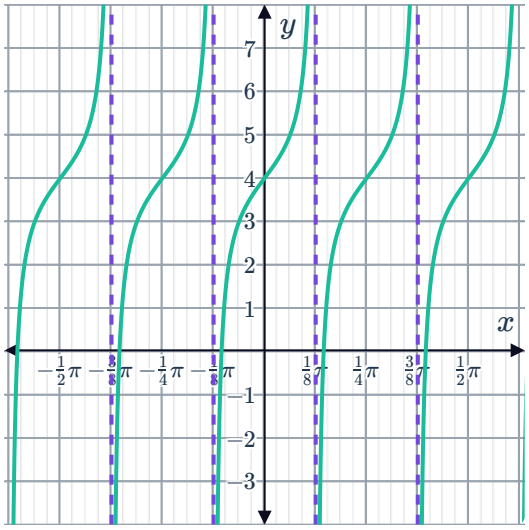
d $y = -\tan\left(x + 2\pi\right)$

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23 The graph $y = \tan x$, as shown, is reflected over the y-axis and then translated vertically 3 units down.
Find the equation of the new graph.



24 Consider the graph of the given tangent function:

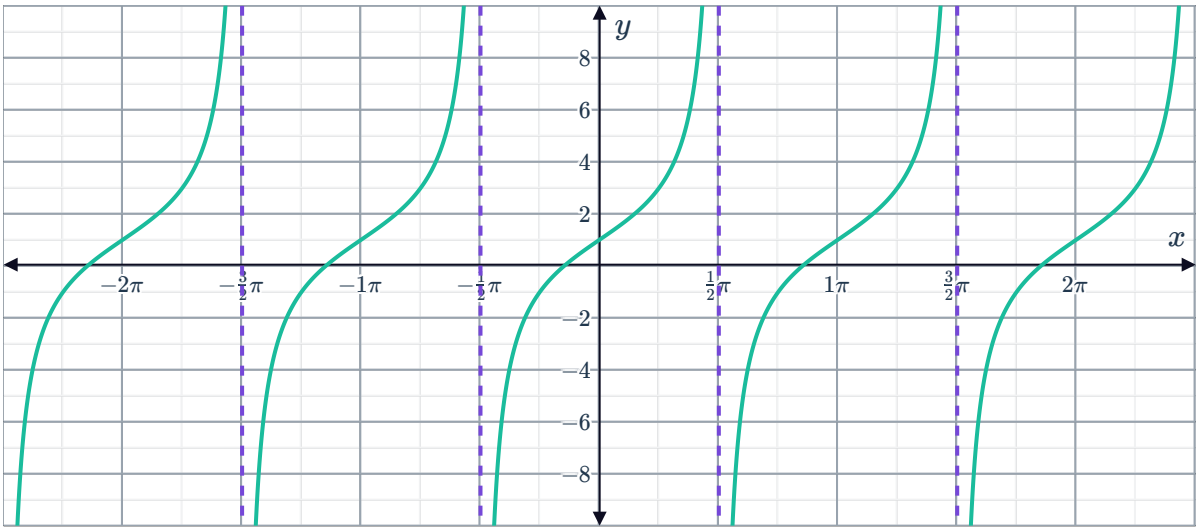


a The given graph is the result of what transformations on $y = \tan x$?

b State the equation of the graphed function.

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25 Consider the given graph of $f(x) = 2\tan x + 1$:



a State the domain of $f(x)$.

b State the range of $f(x)$.

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